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Bioinformatics analysis of Omp19 and Omp25 proteins for designing multi-epitope vaccines against Brucella

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Abstract

Brucellosis is a zoonotic disease caused by Brucella. There is no effective vaccine against human brucellosis. Omp19 and Omp25 are the outer membrane proteins of Brucella. They are widely expressed and highly conserved in Brucella and have high immunogenicity. Herein, we aim to identify multi-epitope vaccine candidates based on Omp19 and Omp25. We analyzed the physicochemical properties and protein structure of Omp19 and Omp25, and predicted the corresponding B cell and T cell epitopes using bioinformatics analysis. Omp19 and Omp25 were composed of 177 amino acids and 213 amino acids, respectively. They were both stable hydrophilic proteins. The instability indices were 44.8 and 23, respectively. The hydrophilicity was –0.1 and –0.317, respectively. In the secondary structure of Omp19 and Omp25 proteins, the α-helix accounted for 12.43% and 23.94%, the β-sheet was 18.64% and 23.47%, the β-turn was 6.78% and 4.23%, and the random coil was 62.15% and 48.36%. Finally, 5 B cell epitopes, 3 Th-cell epitopes and 5 CTL cell epitopes of Omp19 protein, and 4 B cell epitopes, 3 Th-cell epitopes, and 5 CTL cell epitopes of Omp25 protein were selected as vaccine candidates. In conclusion, we obtained potential B cell and T cell epitopes of the Brucella outer membrane Omp19 and Omp25 proteins. This lays the foundation for the further design of multi-epitope vaccine of Brucella.

Abbreviations: MHC = major histocompatibility complex.

Keywords: B cell epitopes, bioinformatics, Brucella, Omp19, Omp25, T cell epitopes

1. Introduction

Brucellosis, caused by Brucella, is one of the most common zoonotic diseases in the world. It severely affects public health in many countries around the world, especially in developing countries in Africa and Central Asia.[1] In China, Xinjiang and Inner Mongolia are provinces with high incidence of brucellosis.[2] The Brucella genus contains at least 10 genetically similar species, the most common being Brucella melitensis.[3] Brucellosis often causes abortion and infertility in mammals, and can be transmitted to humans through the digestive tract and direct contact with broken skin or mucous membranes.[4] Brucella can infect human tissues and organs, causing damage to multiple systems. Although the clinical manifestations of patients are diverse, and the symptoms and signs are atypical, most patients have clinical symptoms of fever, chills, chills, fatigue, night sweats, myalgia, joint pain, headache, backache, and gastrointestinal symptoms.[5]

There are currently no effective vaccines against human brucellosis,[6,7] and vaccines against animal brucellosis have certain shortcomings. For example, the attenuated vaccine S19 may cause orchitis in male animals or abortion in pregnant animals and is harmful to humans.[8] RB51 induces stable immune protection but is resistant to rifampicin, an important antibiotic for the treatment of Brucella.[8] The ev.1 vaccine is resistant to streptomycin in the treatment of brucellosis, is prone to cause abortion in female animals and is harmful to humans.[9,10]

The multi-epitope vaccine is an immunogenic polypeptide designed based on the amino acid sequence of the immunogen,

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The authors have no conflicts of interest to disclose.

The datasets generated during and/or analyzed during the current study are available from the corresponding author on reasonable request.

The study did not include any human or animal experimentation for which ethical approval was required.

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in order to stimulate an effective adaptive immune response with a smaller immunogenic peptide. Some studies have shown that multi-epitope vaccines can produce a more effective immune response to brucella infections.[11] Compared with traditional inactivated vaccines, multi-epitope-based vaccines have many advantages, including lack of pathogens, high safety, ease of production, high stability, and hypo allergenicity.[12] In recent years, vaccine design by bioinformatics analysis of proteins has become an effective method for designing different multi-epitope vaccines.[13,14]

OMP19 and OMP25 are outer membrane proteins widely expressed in Brucella,[15] and are expected to become candidate vaccines for brucellosis.[16] Animal study has shown that oral adjuvant-free vaccine against Omp19 protected against Brucella abortus mucosal infection.[17] Omp25 is highly conserved among Brucella species.[18] Studies have shown that the Omp25 DNA vaccine and recombinant protein vaccine can induce immune response to Brucella melitensis or Brucella abortus in mice.[19,20] Therefore, the epitopes of OMP19 and Omp25 are potential targets for multi-epitope vaccines against Brucella infection.

In this study, we predicted the B cell and T cell epitopes of OMP19 and Omp25 proteins through bioinformatics analysis. The epitope predicted by this study may be used as effective vaccine candidates to prevent brucellosis.

2. Materials and methods

2.1. Ethical statement

The study did not include any human or animal experimentation for which ethical approval was required.

2.2. Amino acid sequence

From the NCBI website (https://www.ncbi.nlm.nih.gov), the amino acid sequence of OMP19 (GenBank: EEZ15319.1) and OMP25 (GenBank: AEF59022.1) of Brucella melitensis was obtained.

2.3. Prediction of physicochemical properties

Prot Param online analysis tool[21] (http://web.expasy.org/protparam/) was used to analyze the physicochemical properties of OMP2b and BCSP31, including amino acid number, theoretical isoelectric point, relative molecular mass, molecular formula, instability coefficient, and hydrophilicity.

2.4. Prediction of protein secondary structure

The secondary structures of OMP19 and OMP25 proteins were analyzed by SOPMA online analysis software[22] (https://npsa-prabi.ibcp.fr/cgi-bin/npsa_automat.pl?page=/NPSA/npsa_sopma.html). The protein features such as α-helix, β-sheet, β-turn and random coil were predicted. The similarity threshold and window width parameters are the default values of 8 and 17, respectively.

2.5. Prediction of protein tertiary structure

The tertiary structure of OMP19 and OMP25 proteins were analyzed by SWISS-MODEL online analysis tool[23] (https://swissmodel.expasy.org/), and illustrated using RasMol 2.7.5.2 software (https://www.rasmol.org/).

2.6. B cell epitope prediction

To improve the accuracy of predicting linear B cell epitopes, we used 4 online analysis tools, including BepiPred1.0[24] (http://www.cbs.dtu.dk/services/BepiPred/), ABCpred[25] (http://www.imtech.res.in/raghava/abcpred/), BCPREDS[26] (http://ailab-projects1.ist.psu.edu:8080/bcpred/), IEDB[27] (http://tools.immuneepitope.org/mhci/).

We first evaluated the length and position of the epitopes predicted by each software. BepiPred 1.0 sets a threshold of 0.35. The threshold for ABCpred is 0.51. BCPRED predicts a linear epitope of 20 amino acids with a default specificity of 75%. The IEDB online software predicts the β-helix, surface accessibility, flexibility and hydrophilicity of the predicted B cell epitopes of Omp19 and Omp25. The default thresholds for β-helix, surface accessibility, flexibility, and hydrophilicity of Omp19 are 1.079, 1, 1.015, and 2.164, respectively; while those of Omp25 are 1.018, 1, 0.995, and 1.663, respectively. We eliminated sequences below the thresholds and selected the overlapping sequences predicted by 4 online tools as promising vaccine candidates.

2.7. T cell epitope prediction

The SYFPEITHI[28] (http://www.syfpeithi.de/bin/MHCServer.dll/Epitope Prediction.htm), NetMHCII 2.3 Server[29] (http://www.cbs.dtu.dk/services/ NetMHCII) and NetMHC 4.0 Server (http://www.cbs.dtu.dk/services/NetMHC/) were used to predict T cell epitopes of OMP19 and OMP25. According to the research by Shen et al,[30] the high frequency HLA alleles in Xinjiang, China, included HLA-A * 1101, A * 0201, and A * 0301, HLA-DRB1 * 0701, DRB1 * 1501, and DRB1 * 0301. Among them, we selected HLA-A * 0201, HLA-A * 0301, and HLA-A * 1101 to predict the corresponding CTL (cytotoxic T lymphocyte) epitope, and the HLA-DRB1 * 0301, HLA-DRB1 * 0701, and HLA-DRB1 * 150 to predict the Th epitope. According to the characteristics of major histocompatibility complex (MHC) molecules, we excluded some sequences that are too long or too short to form epitopes, and then selected overlapping sequences as the final candidate epitopes for Th and CTL based on the high-scoring epitope sequence and percentile ranking.

3. Results

3.1. Amino acid sequence

From the NCBI website (https://www.ncbi.nlm.nih.gov), the amino acid sequence of the outer membrane protein OMP19 (GenBank: EEZ15319.1) and OMP25 (GenBank: AEF59022.1) of Brucella melitensis was obtained, which were as follows:

OMP19: MGISKASLL SLAAAGIVL AGCQSSRLGNL DNVSPPPPPAPVNA VPAGTVQKGN LDSPTQFPNAP STDMSAQSGT QVASLPPASAP DLTPGAVAGV WNASLGGQSC KIATPQTKYG QGYRAGPLRCP GELANLASWA VNGKQLVLYD ANGGTVASLY SSGQGRFDGQT TGGQAVTLSR

OMP25: MRTLKSLVIV SAALLPFSATA FAADAIQEQP PVPAPVEVAP QYSWAGGYT GLYLGYGWNK AKTSTVGSIK PDDWKAGAF AGWNFQQDQ IVYGVEGDAG YSWAKKSKD GLEVKQGFEG SLRARVGYDL NPVMPYLTAG IAGSQIKLNNG LDDESKFRVG WTAGAGLEAK LTDNILGRVEY RYTQYGNKNY DLAGTTVRNKL DTQDFRVGIGYKF.

3.2. Physicochemical properties of OMP19 and OMP25

The physicochemical properties of OMP2b and BCSP31 was analyzed with Prot Param. As shown in Table 1, Omp19 is composed of 177 amino acids, and its isoelectric point is 8.91 with a molecular weight of 17.6KD. Its molecular formula is C764H1224N220O247S5, and its instability index is 44.8, showing that it is a stable protein with a hydrophilicity of –0.1 (hydrophilic protein).

Omp25 is composed of 213 amino acids with an isoelectric point of 8.58 and a molecular weight of 23.1KD. Its molecular formula is C1053H1605N277O311S2, and its instability index is 23, indicating that it is a stable protein with a hydrophilicity of –0.317 (hydrophilic protein).

3.3. Secondary structure of OMP19 and OMP25 proteins

SOPMA analysis showed that in the secondary structure of OMP19 protein contained 12.43% of α-helix, 18.64% of β-sheet, 6.78% of β-turn and 62.15% of random coil (Fig. 1A). For the secondary structure of OMP25 protein, the α-helix was 23.94%, β-sheet was 23.47%, β-turn was 4.23%, and random coil was 48.36% (Fig. 1B). The β turn and the random coils are conducive to the formation of linear B cell epitopes. Therefore, from the secondary structure, we can speculate that both Omp19 and Omp25 contain a large number of regions that could easily form epitopes.

Table 1

The physicochemical properties of Omp19 and omp25.

Items	Omp19	Omp25
Number of amino acids	177	213
Theoretical isoelectric point	8.91	8.58
Molecular mass (Daltons)	17603.75	23185.12
Formula	$C_{764}H_{1224}N_{220}O_{247}S_5$	$C_{1053}H_{1605}N_{277}O_{311}S_2$
Instability index	44.8 (stable protein)	23.00 (stable protein)
Grand average of hydropathicity	–0.100 (hydrophilic protein)	–0.317 (hydrophilic protein)

3.4. Tertiary structure of OMP19 and OMP25

SWISS-MODEL online analysis software and RasMol 2.7.5.2 were used to predict and illustrate the tertiary structure of Omp19 and Omp25 (Fig. 2). The results showed that both Omp19 and Omp25 showed a relatively stable closed-loop structure. The β-turn and random coil in Omp19 and Omp25 were located near the surface of the protein. Therefore, we speculate that these sites have great potential as epitopes for vaccines.

3.5. B cell epitope analysis of OMP19 and OMP25

We used ABCpred, BCPRED, and BepiPred 1.0 to predict the linear B cell epitopes of OMP19 and OMP25. B cell epitopes with higher scores were shown in Table 2. IEDB was used to predict the β-turn, hydrophilicity, flexibility, and surface accessibility of OMP19 and OMP25 B cell epitopes. The default threshold value of the IEDB system was used. The parts above the threshold value were shown in yellow, and parts below were shown in green (Figs. 3 and 4). The higher score represents a higher possibility as a potential epitope. We eliminated sequences below the threshold and selected overlapping sequences predicted by different software as final B cell epitope candidates. Finally, amino acids 32 to 39, 56 to 61, 82 to 87, 117 to 121, and 160 to 168 were used as the final B cell candidate epitopes of Omp19. Amino acids 59 to 71, 93 to 101, 147 to 151, and 182 to 191 were used as the final B cell candidate epitopes of Omp25 (Table 3).

3.6. T cell epitope analysis of OMP19 and OMP25

We used SYFPEITHI, NetMHCII 2.3 Server, and NetMHC 4.0 Server to predict the T cell epitopes of OMP19 and OMP25.

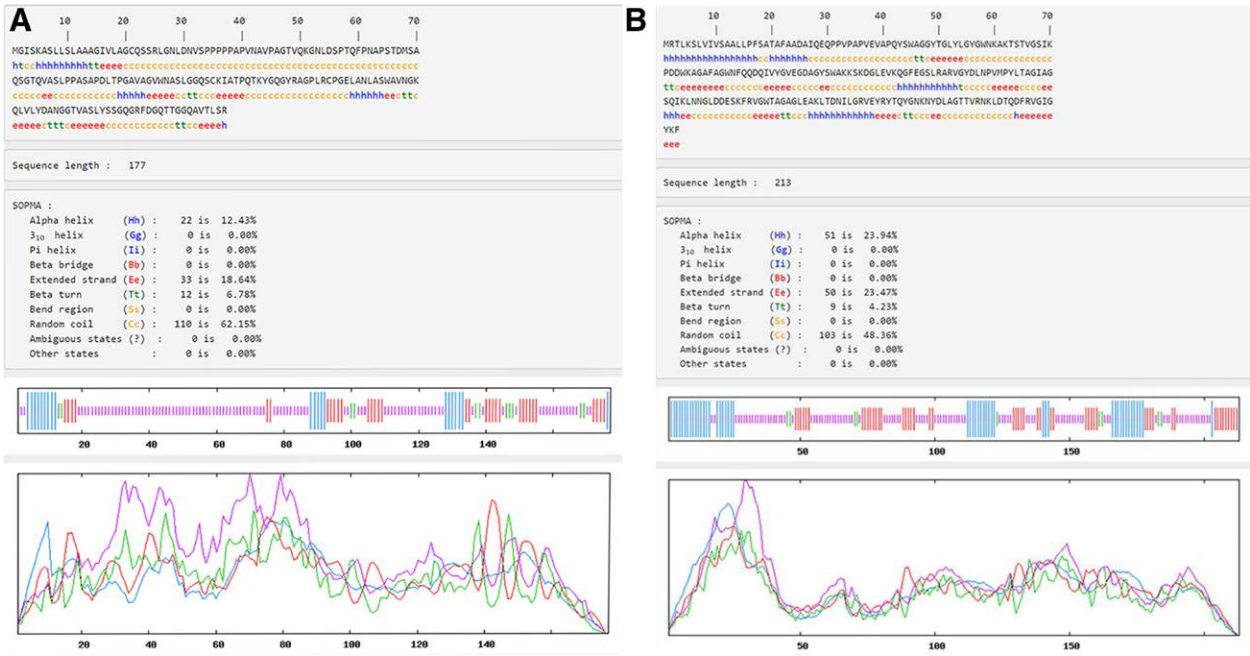


Figure 1. The secondary structure of OMP19 protein and Omp25 protein analyzed by SOPMA. The similarity threshold and window width parameters are the default values of 8 and 17. (A) The secondary structure of Omp19. Alpha helix in the secondary structure of Omp19 protein was 12.43%, β-turn was 6.78%, β-sheet was 18.64%, and random coil was 62.15%. (B) The secondary structure of Omp25. Alpha helix in the secondary structure of Omp25 protein was 23.94%, β-turn was 4.23%, β-sheet was 23.47%, and random coil was 48.36%. The yellow part indicates the epitope position above the threshold, and the green part indicates the epitope position below the threshold.

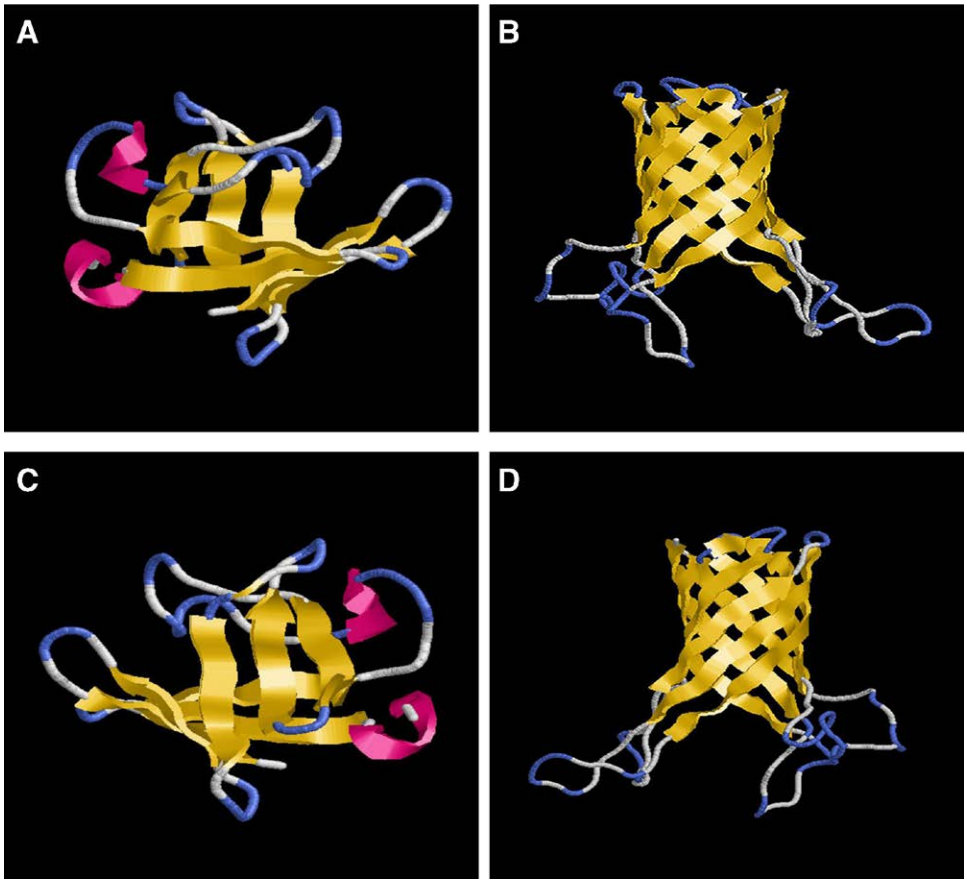


Figure 2. The predicted tertiary structures of the Omp19 and Omp25 proteins. The prediction of the tertiary structures of the Omp19 and Omp25 proteins was performed by the SWISS-MODEL online server and illustrated by the RasMol 2.7.5.2 software. The alpha-helix was marked in red, theβ-sheet in yellow, the random coil in blue, and other residues tags in white. The front views of Omp19 (A) and Omp25 protein, (B) and the reverse views of Omp19, (C) and Omp25 protein, and (D) were shown.

Table 2

The linear B-cell epitopes of OMP19 and OMP25.

	ABC pred prediction			BC Pred prediction			Bepi Pred 1.0 server		
Omp19	Start position	Sequence	Score	Start position	Sequence	Score	Start position	Sequence	Score
	107	IATPQTKYGQGYRAGP	0.9	29	LDNVSPPPPPAPVNAVPAGT	1	29	LDNVSPPPPPAPVNAVP	2
	74	TQVASLPPASAPDLTP	0.9	56	SPTQFPNAPSTDMSAQSGTQ	1	55	DSPTQFPNAPSTDMSA	1.6
	82	ASAPDLTPGAVAGVWN	0.9	80	PPASAPDLTPGAVAGVWNAS	1	156	YSSGQGRFDGQTTGGQAV	1
	46	AGTVQKGNLDSPTQFP	0.9	108	ATPQTKYGQGYRAGPLRCPG	1	105	CKIATPQTKYGQGYRAGPLRC	0.9
	28	NLDNVSPPPPPAPVNA	0.8	155	LYSSGQGRFDGQTTGGQAVT	1	147	ANGGTV	0.7
	160	QGRFDGQTTGGQAVTL	0.8						
	117	GYRAGPLRCPGELANL	0.8						
Omp25	86	QDQIVYGVEGDAGYSW	0.9	26	AIQEQPPVPAPVEVAPQYSW	1	26	AIQEQPPVPAPVEVAPQYSWAGG	1.2
	50	TGLYLGYGWNKAKTST	0.9	87	IVYGVEGDAGYSWAKKSKDG	1	59	NKAKTSTVGSIKPDDWKAG	1
	136	AGIAGSQIKLNNGLDD	0.9	57	GWNKAKTSTVGSIKPDDWKA	0.9	93	VEGDAGYSWAKKSKDGLEVK	0.8
	176	RVEYRYTQYGNKNYDL	0.9				148	GLDDESK	0.8
	65	TVGSIKPDDWKAGAFA	0.9				182	TQYGNKNYDLA	0.7

ABC pred Prediction server score threshold is 0.51. Bepi Pred 1.0 Server score threshold is 0.35.

Specifically, the HLA-A * 1101, HLA-A * 0201, and HLA-A * 0301 were used to predict the CTL epitope. The HLA-DRB1 * 0701, HLA-DRB1 * 1501, and HLA-DRB1 * 0301 were used to predict the Th epitope. We selected high-scoring overlapping sequences as candidate epitopes (Tables 4–7). In the end, we selected amino acids 10 to 18, 87 to 95, 98 to 106, 136 to 144, and 154 to 162 as the Omp19 protein CTL epitopes (Table 3). Amino acids 6 to 20, 15 to 29, and 126 to 154 were selected as the Th cell epitopes of Omp19 protein. The 7 to 15 amino acids, 13 to 21 amino acids, 136 to 144 amino acids, 2 to 10 amino acids, and 168 to 176 amino acids were selected as CTL epitopes of Omp25 protein. Amino acids 11 to 25, 120 to 134, and 154 to 168 were selected as the Th cell epitopes of Omp25 protein.

4. Discussion

Brucella infection poses a serious health threat to herders and livestock in areas with developed animal husbandry,

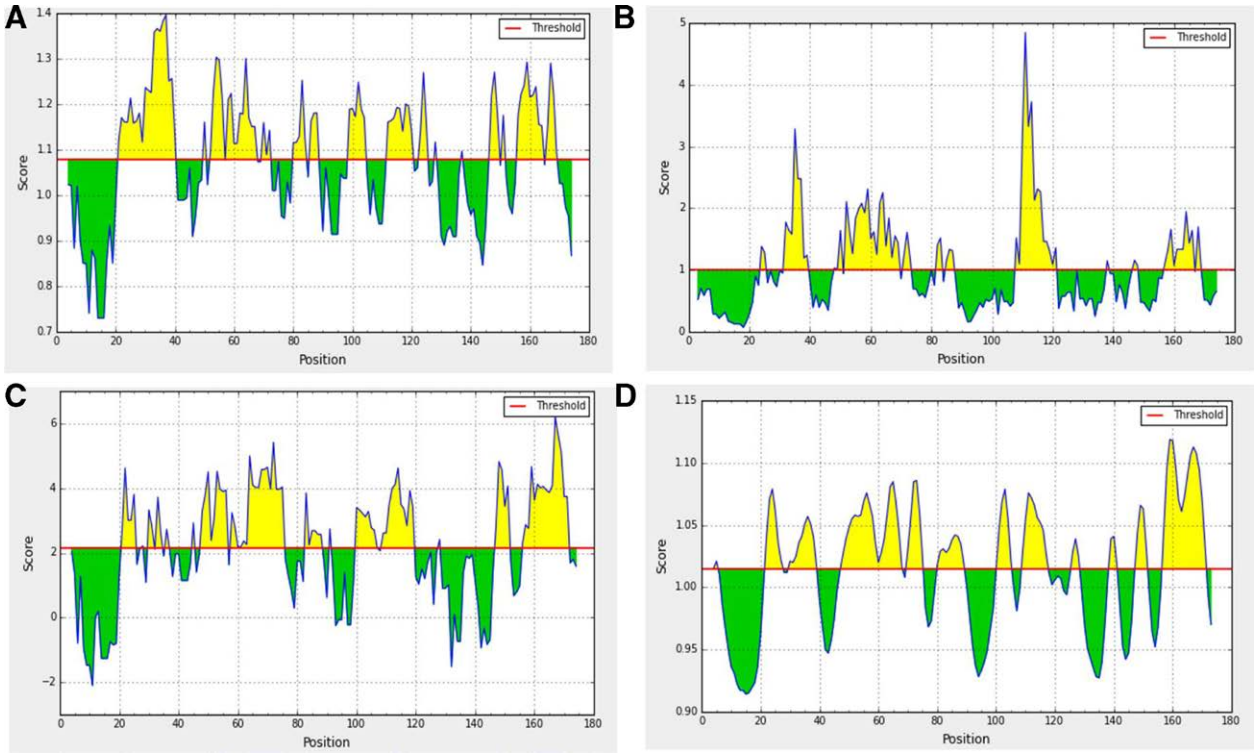


Figure 3. IEDB predicts β-turn, hydrophilicity, flexibility and surface accessibility of Omp19 B cell epitopes. (A) Prediction of β-turn of OMP19 protein by IEDB software. The software is based on the results of Chou and Fasman method to predict the β-turn. The software default threshold is 1.079. (B) Prediction of surface accessibility of Omp19 by IEDB. The software predicts the results based on Emini method. The default value is 1.0. (C) Prediction of Omp19 hydrophilicity by IEDB. Hydrophilic results predicted based on Parker method. The default threshold is 2.164. (D) Prediction of Omp19 flexibility by IEDB. Based on Karplus and Schulz method. The default threshold is 1.015.

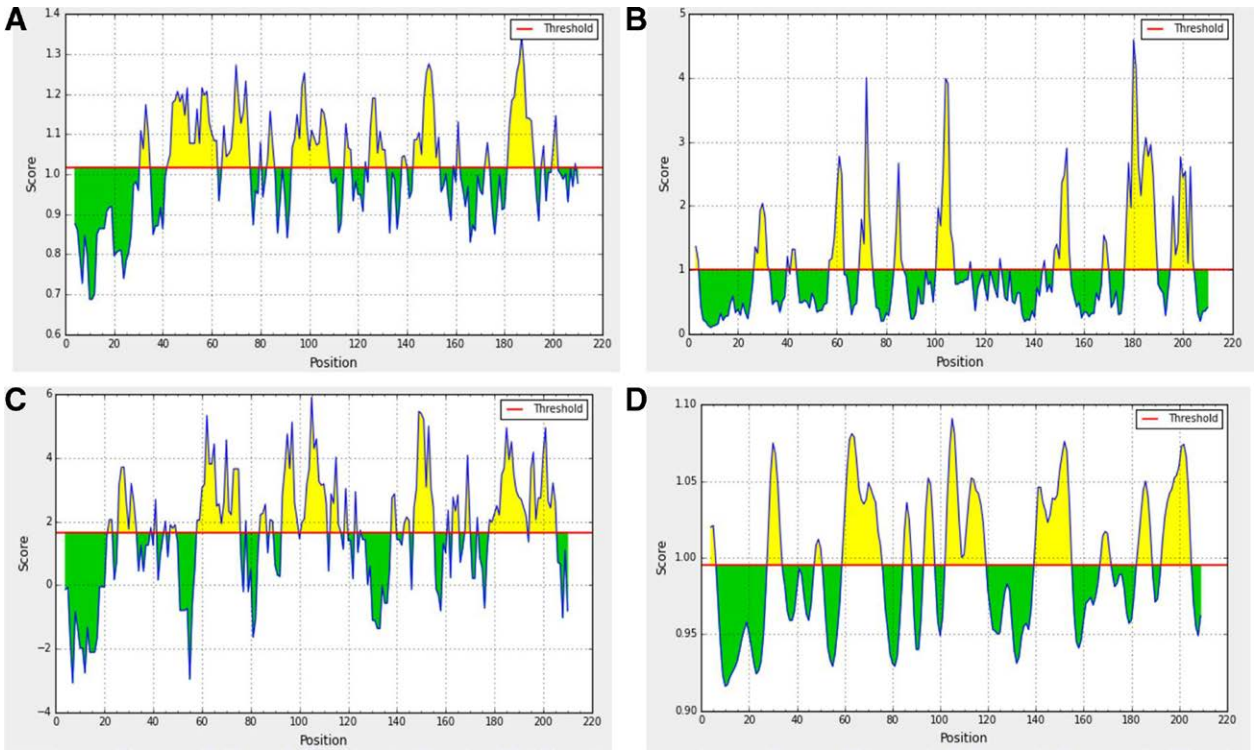


Figure 4. IEDB predicts β-turn, hydrophilicity, flexibility and surface accessibility of Omp25 B cell epitopes. (A) Prediction of β-turn of OMP25 protein by IEDB software. The software is based on the results of Chou and Fasman method to predict the β-turn angle. The software default threshold is 1.018. (B) Prediction of surface accessibility of Omp19 by IEDB. The software predicts the results based on Emini method. The default value is 1.0. (C) Prediction of Omp19 hydrophilicity by IEDB. Hydrophilic results predicted based on Parker method. The default threshold is 1.663. (D) Prediction of Omp19 flexibility by IEDB. Based on Karplus and Schulz method. The default threshold is 0.995.

Table 3

The T- and B cell dominant epitope prediction of the Omp19 and Omp25 protein.

		Omp19		Omp25	
Episode	**Methods**	**Location**	**Sequence**	**Location**	**Sequence**
B cell epitope	Bepi Pred1.0, ABC pred, BCPREDS, IEDB	32–39	VSPPPPPA	59–71	NKAKTSTVGSIKP
		56–61	SPTQFP	93–101	VEGDAGYSW
		82–87	ASAPDL	147–151	NGLDD
		117–121	GYRAG	182–191	TQYGNKNYDL
		160–168	QGRFDGQTT		
CTL	SYFPEITHI, Net MHC 4.0 Server	10–18	SLAAAGIVL	7–15	LVIVSAALL
		87–95	LTPGAVAGV	13–21	ALLPFSATA
		98–106	ASLGGQSCK	136–144	AGIAGSQIK
		136–144	AVNGKQLVL	2–10	RTLKSLVIV
		154–162	SLYSSGQGR	168–176	KLTDNILGR
Th	SYFPEITHI, Net MHCII 2.3	6–20	ASLLSLAAAGIVLAG	11–25	SAALLPFSATAFAAD
		15–29	GIVLAGCQSSRLGNL	120–134	RARVGYDLNPVMPYL
		126–154	PGELANLASWAVNGKQLVLYDANGGTVAS	154–168	KFRVGWTAGAGLEAK

Table 4

Parameters HLA-A*0201 and HLA-A*0301 HLA-A*1101 OMP19 protein T cell epitope prediction.

	Syfpeithi			Net MHC 4.0 server		
HLA	**Position**	**Sequence**	**Score**	**Position**	**Sequence**	**Percentile rank**
HLA_A*0201	87	LTPGAVAGV	25	10	SLAAAGIVL	1.8
	99	SLGGQSCKI	24	8	LLSLAAAGI	2.5
	10	SLAAAGIVL	24	74	TQVASLPPA	2.5
	8	LLSLAAAGI	23	87	LTPGAVAGV	4
	147	ANGGTVASL	22	129	LANLASWAV	4.5
	136	AVNGKQLVL	21	99	SLGGQSCKI	5.5
	11	LAAAGIVLA	20	143	VLYDANGGT	5.5
	24	SRLGNLDNV	20	11	LAAAGIVLA	6.5
	167	TTGGQAVTL	20	90	GAVAGVWNA	6.5
	92	VAGVWNASL	19	7	SLLSLAAAG	7.5
	3	ISKASLLSL	18	83	SAPDLTPGA	8.5
	7	SLLSLAAAG	18	5	KASLLSLAA	13
	17	VLAGCQSSR	17			
HLA_A*0301	43	AVPAGTVQK	34	106	KIATPQTKY	0.5
	86	DLTPGAVAG	23	98	ASLGGQSCK	0.6
	136	AVNGKQLVL	23	154	SLYSSGQGR	0.6
	17	VLAGCQSSR	22	17	VLAGCQSSR	0.9
	98	ASLGGQSCK	22	43	AVPAGTVQK	1
	154	SLYSSGQGR	22	132	LASWAVNGK	1.8
	10	SLAAAGIVL	21	111	QTKYGQGYR	4
	7	SLLSLAAAG	20	116	QGYRAGPLR	5
	16	IVLAGCQSS	20	10	SLAAAGIVL	8.5
	105	CKIATPQTK	20	136	AVNGKQLVL	9.5
				5	KASLLSLAA.	12
HLA-A*1101	43	AVPAGTVQK	28	43	AVPAGTVQK	0.4
	98	ASLGGQSCK	25	98	ASLGGQSCK	0.4
	136	AVNGKQLVL	19	154	SLYSSGQGR	2.5
	154	SLYSSGQGR	18	132	LASWAVNGK	1.3
	94	GVWNASLGG	17	106	KIATPQTKY	3
	23	SSRLGNLDN	17	169	GGQAVTLSR	3
	10	SLAAAGIVL	16	111	QTKYGQGYR	4.5
	17	VLAGCQSSR	16	17	VLAGCQSSR	5
				116	QGYRAGPLR	7
				150	GTVASLYSS	7
				2	GISKASLLS	9
				5	KASLLSLAA	11
				136	AVNGKQLVL	12
				10	SLAAAGIVL	13

The default length of CTL epitope is 9aa.

and results in substantial losses to the farming and pastoral economy.[31] However, there is currently no effective vaccine against Brucella in humans.[32] Studies have shown that using biological information analysis to predict pathogenic B cell and T cell epitopes is a cost-effective method in the design of vaccines.[33–36] Many studies have shown that an effective Brucella vaccine requires humoral and cellular immune-mediated immune responses.[37]

Table 5

Parameters HLA-DRB1*0301 and HLA-DRB1 *0701 HLA-DRB1*1501 OMP19 protein T cell epitope prediction.

	Syfpeithi			Net MHCII 2.3 server		
HLA	Position	Sequence	Score	Position	Sequence	Percentile rank
HLA-DRB1*0301	140	KQLVLYDANGGTVAS	28	132	LASWAVNGKQLVLYD	12
	152	VASLYSSGQGRFDGQ	26	131	NLASWAVNGKQLVLY	14
	89	PGAVAGVWNASLGGQ	25	133	ASWAVNGKQLVLYDA	14
	49	VQKGNLDSPTQFPNA	20	14	AGIVLAGCQSSRLGN	17
	76	VASLPPASAPDLTPG	20	15	GIVLAGCQSSRLGNL	17
	134	SWAVNGKQLVLYDAN	20	134	SWAVNGKQLVLYDAN	19
	15	GIVLAGCQSSRLGNL	18			
HLA-DRB1 *0701	57	PTQFPNAPSTDMSAQ	32	4	SKASLLSLAAAGIVL	2.5
	15	GIVLAGCQSSRLGNL	24	5	KASLLSLAAAGIVLA	3.5
	126	PGELANLASWAVNGK	24	3	ISKASLLSLAAAGIV	4
	6	ASLLSLAAAGIVLAG	22	6	ASLLSLAAAGIVLAG	5
	76	VASLPPASAPDLTPG	22	7	SLLSLAAAGIVLAGC	6.5
	89	PGAVAGVWNASLGGQ	22	8	LLSLAAAGIVLAGCQ	8
	141	QLVLYDANGGTVASL	22			
	152	VASLYSSGQGRFDGQ	20			
	8	LLSLAAAGIVLAGCQ	18			
HLA-DRB1*1501	139	GKQLVLYDANGGTVA	28	138	NGKQLVLYDANGGTV	4.5
	129	LANLASWAVNGKQLV	24	139	GKQLVLYDANGGTVA	4.5
	8	LLSLAAAGIVLAGCQ	20	140	KQLVLYDANGGTVAS	6
	6	ASLLSLAAAGIVLAG	18	5	KASLLSLAAAGIVLA	7.5
	76	VASLPPASAPDLTPG	18	126	PGELANLASWAVNGK	7.5
	89	PGAVAGVWNASLGGQ	18	127	GELANLASWAVNGKQ	7.5
	126	PGELANLASWAVNGK	18	4	SKASLLSLAAAGIVL	8
	140	KQLVLYDANGGTVAS	18	137	VNGKQLVLYDANGGT	8
	141	QLVLYDANGGTVASL	18	6	ASLLSLAAAGIVLAG	11
	15	GIVLAGCQSSRLGNL	18			

The default length of Th epitope is 15aa.

Some proteins from the Brucella cell membrane and cytoplasm have been evaluated as protective antigens in mouse models, including OMP28 (BP26),[38] Omp2b,[39] Cu-Zn superoxide dismutase,[40] and Omp31.[41] Among them, Omp19 is related to bacterial virulence and is widely expressed in Brucella.[42] Pasquevich et al[15] found that the lipidated Omp19 protein had a significant protective effect against Brucella abortus infection, and had a similar level of protection induced by the S19 vaccine. And in their subsequent research,[17] they found that the purified nonlipidated Omp19 protein elicited a protective immune response in mice. We speculate that a vaccine containing the Omp19 epitope may generate a protective immune response against multiple Brucella infections. Omp25 is highly conserved among Brucella species,[43] and can inhibit the production of tumor necrosis factor alpha by human macrophages infected with Brucella Suis.[44] Studies by Goel et al[20,45] and Paul et al[46] showed that the immune response produced by recombinant Omp25 in mice protected the infection of Brucella melitensis and Brucella abortus. These studies suggest that Omp19 and Omp25 are effective candidates for human brucellosis vaccines. In this study, we predicted the potential B cell and T cell epitopes of Omp19 and Omp25. According to the physicochemical properties predicted by Prot Param, Omp19 and Omp25 were both highly hydrophilic proteins. Their isoelectric points were 8.91 and 8.58, respectively. The prediction of the instability index showed that both of them were stable. These biochemical parameters will provide references for the extraction and purification of Omp19 and Omp25.

At present, the commonly used methods to predict the secondary structure of proteins are PHD, SOPMA, GOR, etc. Among them, SOPMA uses an optimization scheme and can combine several independent secondary structure predictions into 1 consistent result.[22] Using SOPMA analysis, we analyzed the secondary structures of OMP19 and OMP25. Protein secondary structure mainly includes α-helix, β-sheet, β-turn and random coil. The α-helix and β-sheet play an important role in maintaining the stability of the secondary structure of the protein, but they are mostly located inside the protein. Thus, it is more difficult for them to form epitopes. However, β-turn regions and random coil regions are mostly found on the surface of proteins. Therefore, β-turn regions and random coil regions are conductive to the formation of linear B cell epitope regions. The results showed that in the secondary structure of Omp19, α-helix accounted for 12.43%, β-sheet accounted for 18.64%, and, β-turn and random coil accounted for 6.78% and 62.15%, respectively. In the secondary structure of OMP25, α-helix accounted for 23.94%, β-sheet accounted for 23.47%, and, β-turn and random coil accounted for 4.23% and 48.36%, respectively. From the secondary structure, we can speculate that both Omp19 and Omp25 contain regions that easily form epitopes.

In addition, we use SWISS-MODEL, which is a commonly used protein tertiary structure prediction tool,[23] to predict the tertiary structure. In Rasmol software, we choose the Cartoons mode as the representation form of Omp19 and Omp25. Cartoons mode is mainly used to display some special structures of macromolecules, such as alpha helix and beta fold. We found that its β-turns and random coil structures were located on the outside of the protein, suggesting that these structures are prone to form B cell linear epitopes.

Humoral immunity plays an important role in the initial stages of the Brucella infection. But when Brucella resides inside the cell, the effective immune response against Brucella is mainly cellular immunity. In addition, the production of interferon-γ by helper T cells 1 and CTL cell-mediated responses are key mediators of protective immunity against Brucella infection.[17] Therefore, we analyzed the B-, Th-, and CTL-cell epitopes of Omp19 and Omp25.

B cell epitopes play a vital role in vaccine design, immunodiagnostic design, and immunogen design for antibody production.[47] In this study, we analyzed the linear B cell epitopes of Omp19 and Omp25. To improve the accuracy of predicting

Table 6

Parameters HLA_A*0201 and HLA_A*0301 HLA-A*1101 OMP25 protein T cell epitope prediction.

	Syfpeithi			Net MHC 4.0 server		
	Position	Sequence	Score	Position	Sequence	Percentile rank
HLA_A*0201	6	SLVIVSAAL	25	13	ALLPFSATA	0.25
	13	ALLPFSATA	25	199	KLDTQDFRV	0.25
	137	GIAGSQIKL	25	130	VMPYLTAGI	1.2
	126	DLNPVMPYL	23	85	QQDQIVYGV	1.7
	2	RTLKSLVIV	22	6	SLVIVSAAL	1.8
	7	LVIVSAALL	20	133	YLTAGIAGS	1.8
	133	YLTAGIAGS	20	53	YLGYGWNKA	2
	44	SWAGGYTGL	19	126	DLNPVMPYL	3
	168	KLTDNILGR	17	2	RTLKSLVIV	3.5
	190	DLAGTTVRN	17			
	172	NILGRVEYR	16			
	118	SLRARVGYD	16			
HLA_A*0301	176	RVEYRYTQY	26	62	KTSTVGSIK	0.15
	13	ALLPFSATA	25	54	LGYGWNKAK	1.7
	9	IVSAALLPF	21	168	KLTDNILGR	1.9
	104	KSKDGLEVK	21	104	KSKDGLEVK	2.5
	168	KLTDNILGR	21	191	LAGTTVRNK	3.5
	136	AGIAGSQIK	20	117	GSLRARVGY	4.5
	7	LVIVSAALL	19	160	TAGAGLEAK	4.5
	118	SLRARVGYD	19	52	LYLGYGWNK	5.5
	172	NILGRVEYR	18	136	AGIAGSQIK	5
	6	SLVIVSAAL	17	9	IVSAALLPF	6
	133	YLTAGIAGS	17	2	RTLKSLVIV	7.5
	190	DLAGTTVRN	17	67	GSIKPDDWK	7.5
				13	ALLPFSATA	8.5
HLA-A*1101	67	GSIKPDDWK	24	62	KTSTVGSIK	0.15
	104	KSKDGLEVK	23	160	TAGAGLEAK	1.4
	168	KLTDNILGR	21	67	GSIKPDDWK	1.8
	148	GLDDESKFR	19	52	LYLGYGWNK	2.5
	172	NILGRVEYR	17	117	GSLRARVGY	3
	2	RTLKSLVIV	15	136	AGIAGSQIK	3
	7	LVIVSAALL	15	168	KLTDNILGR	3.5
	9	IVSAALLPF	15	112	KQGFEGSLR	5
	137	GIAGSQIKL	15	9	IVSAALLPF	5
				10	VSAALLPFS	7.5
				2	RTLKSLVIV	8.5

The default length of CTL epitope is 9aa.

epitopes, we used Bepi Pred, ABC pred, BCPREDS, and IEDB to analyze linear B cell epitopes. Bepi Pred is a method for obtaining B cell epitopes based on the combination of a hidden Markov model and the optimal propensity scale method.[24,48] The score threshold is set to the default value (0.35). ABC Pred is a prediction method using a recurrent neural network,[25] with amino acid length and threshold set to 20 and 0.51, respectively. BC Pred is based on the subsequence support vector machine theory.[49] The default analysis sequence length is 20 amino acid sequences, and the default specificity is 75%. The IEDB predicts β-turn, hydrophilicity, flexibility and surface accessibility parameters of B cell epitopes. In addition, regions with strong hydrophilicity, good accessibility and plasticity are also the dominant regions for forming epitopes.[50] Comprehensively analyzing the results of the 4 analysis softwares, we eliminated sequences below the threshold and selected overlapping sequences with higher scores as the final B cell candidate epitopes. In the end, we used amino acids 32 to 39, 56 to 61, 82 to 87, 117 to 121, and 160 to 168 as the final B cell candidate epitopes of Omp19. Amino acid 59 to 71, amino acid 93 to 101, amino acid 147 to 151, amino acid 182 to 191 were used as the final B cell candidate epitopes of Omp25.

T cell epitopes are short protein peptides that bind to MHC molecules and are recognized by T cell receptors. T cell receptor is able to recognize the antigen when the surface of the antigen-presenting cell is bound to the MHC molecule. T cell epitopes are presented by class I (MHC I) and II (MHC II) MHC molecules, which are recognized by 2 different T cell subsets, CD8 + and CD4 + T cells, respectively. CD8 + T cells become CTLs after CD8 epitope recognition. Meanwhile, activated CD4 + T cells can differentiate into helper (Th) or regulatory (Treg) T cells.[28] It isshown that,[30] among the Uighur population in Xinjiang, China, HLA-A * 1101 (13.46%), A * 0201 (12.50%), A * 0301 (10.10%), and HLA-DRB1 * 0701 (16.35%), DRB1 * 1501 (8.65%), DRB1 * 0301 (7.69%) had high frequency distribution. Therefore, we use HLA-A * 1101, HLA-A * 0201, and HLA-A * 0301 to predict the corresponding CTL epitope, and HLA-DRB1 * 0701, HLA-DRB1 * 1501, and HLA-DRB1 * 0301 to predict the Th epitope. Bit. We use SYFPEITHI and NetMHCII to predict T cell epitopes of OMP19 and OMP25 proteins. SYFPEITHI is based on the principle of the motif matrix,[51] and NetMHC is a model based on artificial neural networks.[29,52] MHC I molecules can only bind relatively short peptides of 9 to 11 amino acids, and MHC II molecules can bind longer peptides.[28] In the end, we selected amino acids 10 to 18, 87 to 95, 98 to 106, 136 to 144, and 154 to 162 as the CTL epitopes of Omp19 protein. Amino acids 6 to 20, 15 to 29, and 126 to 154 were determined as the Th cell epitopes of Omp19 protein. The 7 to 15 amino acids, 13 to 21 amino acids, 136 to 144 amino acids, 2 to 10 amino acids, and 168 to 176 amino acids were selected as CTL epitopes of Omp25 protein. Amino acids 11 to 25, 120 to 134, and 154 to 168 were determined as the Th cell epitopes of Omp25 protein.

Table 7

Parameters HLA-DRB1*0301 and HLA-DRB1 *0701 HLA-DRB1*1501 OMP25 protein T cell epitope prediction.

	Syfpeithi			Net MHCII 2.3 server		
	Position	Sequence	Score	Position	Sequence	Percentile rank
HLA-DRB1*0301	120	RARVGYDLNPVMPYL	27	120	RARVGYDLNPVMPYL	3.5
	170	TDNILGRVEYRYTQY	26	121	ARVGYDLNPVMPYLT	4
	144	KLNNGLDDESKFRVG	26	122	RVGYDLNPVMPYLTA	4
	11	SAALLPFSATAFAAD	23	119	LRARVGYDLNPVMPY	5.5
	4	LKSLVIVSAALLPFS	22			
	154	KFRVGWTAGAGLEAK	19			
HLA-DRB1 *0701	55	GYGWNKAKTSTVGSI	32	199	KLDTQDFRVGIGYKF	1.5
	12	AALLPFSATAFAADA	26	1	MRTLKSLVIVSAALL	4.5
	127	LNPVMPYLTAGIAGS	24	151	DESKFRVGWTAGAGL	5
	154	KFRVGWTAGAGLEAK	24	152	ESKFRVGWTAGAGLE	5.5
	112	KQGFEGSLRARVGYD	24	153	SKFRVGWTAGAGLEA	5.5
	3	TLKSLVIVSAALLPF	22	154	KFRVGWTAGAGLEAK	6.5
	120	RARVGYDLNPVMPYL	22	3	TLKSLVIVSAALLPF	7
	152	ESKFRVGWTAGAGLE	22	4	LKSLVIVSAALLPFS	8
				155	FRVGWTAGAGLEAKL	9
				53	YLGYGWNKAKTSTVG	10
HLA-DRB1*1501	127	LNPVMPYLTAGIAGS	28	10	VSAALLPFSATAFAA	4
	11	SAALLPFSATAFAAD	24	11	SAALLPFSATAFAAD	4
	24	ADAIQEQPPVPAPVE	24	12	AALLPFSATAFAADA	5
	43	YSWAGGYTGLY L G Y G	24	9	IVSAALLPFSATAFA	5
	5	KSLVIVSAALLPFSA	24	1	MRTLKSLV IVSAALL	5.5
	39	QPPVPAPVEVAPQYS	20	127	LNPVMPYLTAGIAGS	9
	30	QPPVPAPVEVAPQYS	20	8	VIVSAALLPFSATAF	9.5
	4	LKSLVIVSAALLPFS	18	128	NPVMPYLTAGIAGSQ	11
	154	KFRVGWTAGAGLEAK	18			

The default length of Th epitope is 15aa.

With the rapid development of computational methods and the production of large amount of experimental immunological data, a new field of research called immunoinformatics has arisen. Immunoinformatics focuses on the design and development of computational and mathematical models for the analysis of the immune system through immunological data. Safavi et al[53] screened out the high immunogenic region of TMEM31 antigen according to the CTL epitopes and MHC binding affinity. They designed the multi-epitope DNA and peptide cancer vaccines consisting of the most immunodominant epitopes of ACRBP and SYCP1 antigens by using bioinformatic tools. In another study by Safavi et al[54], prophylactic combination immunization with multi-epitope DNA and polypeptide cancer vaccines activated the immune system against breast cancer in the animal model. Thus, their findings fully demonstrate the feasibility of bioinformatics in predicting epitopes of antigens and developing vaccines. In future studies, we will develop vaccines based on the prediced epitopes.

5. Conclusion

In this study, we predicted the physicochemical properties of Brucella outer membrane Omp19 and Omp25 protein. We used different methods and parameters to predict the B cell and T cell epitopes of Omp19 and Omp25 protein. The potential B cell and T cell epitopes of Omp19 and Omp25 protein were identified. These predicted B cell and T cell epitopes provide basis for the design of Brucella multi-epitope vaccines.

Author contributions

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